Rajan Dhingani
(PMP, TuV FSC L2 Railway EN 5012x, Automotive L1 ISO 26262, CSPO, CSM, CPRE-FL, ISTQB-TM, ISTQB-Agile, ISTQB-FL, Six-Sigma, INCOSE)

Project Manager | Product Manager | Function Safety & Security | Tech Mng

A highly skilled Technical Manager with over 16 years of experience in system software automation, engineering design, testing, commissioning, and erection of railway and automotive systems. Proficient in managing large teams and projects while ensuring safety and compliance with international standards. Extensive expertise in embedded systems, IoT protocols, and cybersecurity, with a strong focus on functional safety across various industries, including IoT, Railways, Automotive, Power Electronics, Semiconductor, Industrial 4.0, Automation, etc.

CONTACT

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Current Location: Hyderabad India

CORE COMPETENCIES

- Delivery Management
- Deployment Management
- Technical Management
- Stakeholder Management
- Requirement Management
- Budgeting & Cost Control
- Schedule Management
- Function Safety Management
- Project Management
- Product Management
- Product Safety Management
- Cyber Security Management
- Quality Management
- Risk Management
- People Management
- Talent Management
- Supply chain Management
- Procurement Management
- Proposals and RFP, RFQ
- Technology Management
- Innovation Management
- NPD. AI/ML, Gen-AI
- FAI, FAT, SAT, PLM
- A/B Test, EACAD, FMEA

Innovation

- 2 pattern filed for the SW Methodology
- Pattern filed for the Hybrid Green Field Energy System & Solar roof top Electrical bus

White paper

- Satellite Communication for the Railway System
- Cyber Security in Train Control System
- Future Energy Source BEV vs FCEV

Professional Summary:

- **Product, Portfolio, Program and Project Management:** Life Cycle, Delivery, Deployment, NPD, Technical, Stakeholder, Requirement, Budgeting, Cost Control, Function Safety, Product Safety, Security, Quality, Risk, People, cross-functional teams, managing large-scale projects, and ensuring delivery within KPI, scope, time, and budget constraints.
- **IoT & Automation:** IoT Protocols/Platforms (Node-RED, OCPP, Vector vSECC, Grafana, Kafka Lambda, WSN BLE, ZigBee, UART, HART), Communication Protocols (CAN, Ethernet/IP, Modbus), Tools (CANoe, vTeststudio, AutoSAR, dSpace, Jenkins).
- **Embedded IoT:** Design and integration of IoT solutions for railways and automotive systems, including communication protocols (CAN, Modbus, ProfiNet, MVB, WTB, OCPP, MQTT, Web Socket, SATCOM).
- **Railway TCMS:** Design, development, and implementation of TCMS for locomotives, LRV, and metro trains, Mono rails and Signaling
- **Hydrogen & Battery Traction:** Development of hydrogen train control systems and battery traction systems, including high-speed battery charging monitoring and control.
- **Functional Safety & Cybersecurity:** Expertise in EN 50126, EN 50128, EN 50129, ISO 26262, and IEC 61508, focusing on safety-critical systems and IEC 62443 & ISO 21434 cybersecurity.
- **System & Software Engineering:** ASPICE, V-Model and Waterfall Model, Agile, Scrum, KANBAN, LEAN, SIX Sigma processes for software implementation and testing, including SIL2/SIL4, ASIL-C safety documentation for System, Schematics, Software and testing.

Education

Degree	Specialization	University	%/ CGPA	Started year	Completed year
M.Tech	SW System IOT Embedded	(BITS-Pilani) WILP	8.68	Jan-2023	Dec-2024
B.E.	Electronics and Communication	Gujarat University	68.9%	July-2005	Jul-2009
H.S.C.	Science	G.H.S.E.B.	63.3%	May-2004	Mar-2005
S.S.C	English	G.H.S.E.B.	74.3%	May-2002	Mar-2003

Professional Certificate

- TuV CENELEC Function Safety Engineer Level 2 Railway RAMS (EN5012x)
- TuV CENELEC Function Safety Engineer Level 1 Railway RAMS (EN5012x)

Railway and Automotive Platform

- PHA, FMEA, FTA, HARA
- Cyber Security TARA
- Concept Requirements
- Customer Requirements
- Electrical Schematics
- System Requirements
- System Architecture
- SW requirements, SW Architecture, SW Component
- SW Component, integration, Validation test.
- SW-HW integration
- Product test, FAI, Type Test
- Homologation, Vehicle Test, Assessment, System Test

IoT Protocols / Platforms

- Node-RED, OCPP 2.0.1
- Vector vSECC vCharM
- Grafana, Kafka, Lambda
- WSN, BLE, ZigBee, HART
- RF, NFC, SATCOM
- MQTT, XMPP, CoAP, gRPC
- RT-ETH, SerCos, PKI
- OPC UA, OPC DA, RESTful
- AZURE IoT Hub, AWS IoT Core, GCP IoT, Jenkins
- Cloud (IaaS, PaaS. SaaS)

Communication Protocols:

- CAN, CANOpen, TTCAN
- CANASP, J1939, UDS
- Flex, LIN, Field Bus
- Ethernet/IP (TCP/UDP)
- IPTCOM, Ethernet TRDP
- ETB, WTB, MVB, TRDP
- ProfiNet, Profibus, DeviceNet
- Modbus TPC/RTU, RS-422 DeviceNet, RS-485, RS-232,
- SPI, I2C, UART, GPIO
- Protocol Tunneling and Encapsulation

TOOLs / Platform:

- Vector CANoe, vTeststudio
- Jenkins, Maven, Selenium
- MATLAB, Simulink, VT
- CodeSys, TIA, Rockwell
- I-CON_L, RS View, Selectron CAP1131, LabVIEW, Selectron Tool Chain
- PTC, EA, Magic Draw

- TuV CENELEC Function Safety Engineer Level1 Automotive ISO26262
- REB Certified Professional for Requirements Engineering - Foundation Level (CPRE-FL)
- International Software Testing Qualifications Board (ISTQB)- Test Manager (ISTQB-TM), ISTQB-Agile, ISTQB-FL
- Certified Scrum Master (CSM)
- Certified Scrum Product Owner (CSPO)
- Certified Executive Leadership Coaching Program
- Workshop on ‘Charging Infrastructure Technology Requirements & Interoperability’ by ARAI
- Certified Function Safety TS250 & TS255 trainer by Bombardier
- Project Management Professional PMP

Cyient (R&D TCMS Architecture) OCT-2025 to till

Key Responsibilities

- Lead a 15+ member team managing KPIs, scope, quality, and domain-level strategy for TCMS, Traction, HVAC, Doors, energy storage and traction projects.
- Oversaw product and platform development focusing on EMU TCMS, traction batteries, and high-speed charging systems.
- Developed and tested Railway TCMS requirements for traction and auxiliary railway applications, aligned with EN 50126/28/29 and ISO 26262.
- Implemented R&D roadmap for green energy solutions, fostering innovation and continuous improvement.
- Mentored and coached cross-functional teams, P&L, Quality

Amber Sidwal (R&D Electronics AGM) AUG-2025 to OCT-2025

Key Responsibilities

- Lead a 40+ member team managing KPIs, scope, quality, and domain-level strategy for TCMS, HVAC, Doors, energy storage and traction projects.
- Oversaw product and platform development focusing on ESS, traction batteries, and high-speed charging systems.
- Developed and tested ESS requirements for traction and auxiliary railway applications, aligned with EN 50126/28/29 and ISO 26262.
- Implemented R&D roadmap for green energy solutions, fostering innovation and continuous improvement.
- Mentored and coached cross-functional teams

Knorr-Bremse (Sr. R&D Team Lead) Sep-2020 to Aug-2025

Key Responsibilities

- Lead 30+ member team, managing KPIs, scope, quality, time, and efforts for projects.
- Managed development and test environments for product & platform development, focusing on safety-critical railway and automotive systems.
- Implemented V-Model for SW development and testing, ensuring SIL2/SIL1 compliance for TCMS & traction controllers with New Product Development NPD.
- TCMS and traction control systems for SIL4, SIL3, SIL2, SIL1 and SIL0 Railway projects & QM, ASIL-C, ASIL-B, ASIL-A, QM for Automotive

- DOORS-NG, DOORS
- JIRA, SVN, GIT-HUB, JAMA
- Polarion, Pro.File, PTC
- ECAD, EPLAN, CADSTAR
- AutoCAD, CREO, Solid
- SharePoint, Power BI
- Code Composer, Doxygen
- CANDB++, DBC, GDB
- Polarion, Profile,
- Primavera p6, PSRG, PENTA
- RTOS VxWork, vSphere, Linux, Android, WinOS
- QT, QNX, Hyper-V
- VM, VMWare, Virtual Box

Language Skill / Platform:

- Embedded C, C, C++
- Python, Perl, Pascal
- CSS, HTML, JAVA, JS
- IEC-61131, FBD, IS, ST

Overseas Experience

- KIEPE Electrical Dusseldorf Germany
  E-Bus Control system and SW release test for the City of Paris & LRV Project TCMS and Traction System Test (3 months)
- Bombardier London Derby UK for LOT, EAA, SWL, WML, Cross Rail TCMS and Sub System Commissioning. (1 year)
- Bombardier Mannheim Germany for Locomotive TRAXX-3 Project TCMS Commissioning (3 Months)
- Bombardier Zurich Switzerland for Locomotive TRAXX-3 Project TCMS Commissioning (2 Weeks)
- Rockwell Automation Canada AB Controller, LV and MV Power Electronics Control and Monitoring system SAT and FAT (3 Months)
- Hitachi-Hirel, Taiwan, Cooled rolling mill control and monitoring system FAT and SAT (2 Months)

Languages

- English
- Hindi
- Gujarati

- Developed and tested SW system requirements and test, environment for safety-critical railway applications using V-Model, ensuring compliance with EN 50126, EN 50128, EN 50129, and ISO 26262 standards.
- Spearheaded projects including the Dortmund TCMS validation, City of Cologne TCMS and traction system, BMS, Signaling, ATP, ATO, ETCS, ERTMS, CBCT and the K-Charger CCS Charger, Melbourne B2 Control and Traction system
- City-K Charger High speed charger AC/DC, DC/DC Product Development and Testing
- Implemented Python scripts for automated code generation and testing templates for SIL2 components.
- Product development, test, integration and deployment using Jenkins CI/CD, Azure DevOps

Bombardier (Application SW Engineer) Jul-2017 - Sep-2020

Key Responsibilities

- Lead multi-national teams across Germany, UK, and Switzerland on various railway automation projects with team size 12 members.
- Developed SW architecture and interface documents for TCMS functions, traction, Doors, Brake, PIS, JRU, Signaling, ATP, ATO, ETCS, ERTMS, CBCT safety standards compliance.
- Engineered TCMS software for railway functions, including door control, brake systems, and power supply controllers, using C, C++, and FBD.
- Design develop and testing for the SIL controllers CCU-S/ CCU-S2
- Developed safety documentation for SIL1, SIL2, and SIL3 applications, leading base platform and application projects like the Aventra Family and TRAXX-3 family, Talent, Zefiro & QNGR projects.
- Collaborated with global teams across Germany, UK, Switzerland, and India for system integration and testing, Jenkins, CI/CD, AZURE,

Rockwell Automation (Project Engineer) May-2015- Jul-2017

Key Responsibilities

- Product Management controller and drives controller projects, leading teams across India, Canada, and the USA.
- Designed and implemented control and automation systems using Rockwell controllers and HMI, SCADA solutions.
- Designed and tested controllers and TCU, DPU, drive systems for industrial automation projects, focusing on MV and LV drives.
- Prototype design & Test and Certification for Power Electronics and Semiconductor products
- Lead the R&D engineering team for the power controllers of MV Drives and Starters, including synchronization transfer projects and control board designs with team size 8 members.

Hitachi-Hirel (Automation Engineer) May-2012 to May-2015

Key Responsibilities

- Developed Embedded SW design and testing solutions for TCU, controller, VFD Controllers, and HMI, SCADA systems.
- Lead engineering projects across multiple countries, focusing on high-performance drive setups.
- Power Electronics Product development and testing specially Gate driver cards, Communication protocols and Display Designed and commissioned

- German (Beginner)
- Marathi

VFD panels, Control panels, and Monitoring and Control systems for various industrial applications for FAT and SAT.

- Developed test bench automation systems for MV and LV drives and Traction system and ensuring compliance with CE, IEC, and UL certifications

MICT DP World (Engineer) Sep-2011 - May-2012

Key Responsibilities

- Lead the various small and big Project with 3 to 8 members of team including colleagues from India, Dubai, Singapore.
- Installation, Programming, Preventing Maintenance & attaining Breakdown of Load Cells, Electronics Controllers, Encoder, RTD, Thermistor, Position sensor, Proximity Switch, Pressure gauges, Power meters, HT line Protection relay etc.
- Managed the installation, programming, and maintenance of automation systems for container cranes, controllers and HIM programing and testing.

Electrotherm (Automation Engineer) Aug-2009 - Sep-2011

Key Responsibilities

- Lead the various small and big Project with 3 to 8 members of team including colleagues
- Lead the development, Testing, engineering, commissioning of Blast Furnace, Sinter and ESP projects development and testing

Here, I declared that the above-mentioned information is correct up to my knowledge and I bear the sole responsibility for the authenticity.

Rajan Dhingani